Parkinson's Disease Detection from Voice Biomarkers

A Custom MLP with Focal Loss and Ablation Studies

CS273P — Machine Learning Final Project

Wonjae Lee, Jichuan Wu, Leo Zheng

Abstract

We implement a custom multi-layer perceptron (MLP) in PyTorch for detecting Parkinson's disease from 22 voice biomarker features. We employ Focal Loss to handle the 75:25 class imbalance and enforce subject-level cross-validation to prevent data leakage. Systematic ablation studies over loss functions, architecture depth/width, and dropout rates reveal that: (1) Focal Loss and standard BCE perform comparably on F1, but differ on MCC; (2) shallower architectures match or outperform deeper ones on this small dataset; and (3) light dropout ($p \approx 0.1$) provides the best regularization. Our honest subject-level evaluation yields F1=0.893 and MCC=0.448, substantially below the >90% accuracy commonly reported without proper subject splitting.

1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects approximately 10 million people worldwide. Clinical diagnosis through the UPDRS remains subjective and expensive. Voice analysis offers a promising non-invasive alternative: PD causes measurable changes in vocal fold vibration patterns, including increased jitter, shimmer, and altered nonlinear dynamics [1].

This project implements a deep learning classifier for PD detection from voice measurements. We focus on three aspects beyond simple classification: (1) handling class imbalance via Focal Loss [2], (2) preventing data leakage via subject-level cross-validation, and (3) understanding the sensitivity of results to design choices through ablation studies over loss functions, architectures, and regularization.

2 Related Work

Little et al. [1] introduced this dataset and achieved ∼91% accuracy with SVMs. Subsequent work applied random forests and gradient boosting, reporting 85–95% accuracy. However, most studies use random train-test splits that leak subject information (each subject contributes ∼6 recordings) and report only accuracy despite 75% positive-class prevalence. Our work addresses both issues with subject-level splitting and imbalance-aware metrics.

3 Dataset

The UCI Parkinson's dataset [1] contains 195 voice recordings from 32 subjects (24 PD, 8 healthy). Each recording has 22 features: fundamental frequency measures (3), jitter (5), shimmer (6), harmonics-to-noise ratios (2), and nonlinear dynamical measures (6). The class distribution is imbalanced: 147 PD (75.4%) vs. 48 healthy (24.6%).

4 Model Design

4.1 MLP Architecture

We implement a custom MLP in PyTorch with configurable depth, width, and dropout. Each hidden layer consists of:

$$\mathbf{h}_{l+1} = \text{Dropout}\Big(\text{ReLU}\big(\text{BatchNorm}(\mathbf{W}_l\mathbf{h}_l + \mathbf{b}_l)\big)\Big) \tag{1}$$

The final layer is a single linear unit producing a logit: $\hat{y} = \mathbf{w}^\top \mathbf{h}_L + b$, passed through sigmoid for prediction. Our baseline configuration uses hidden layers $[64, 32]$ with dropout $p = 0.3$.

4.2 Focal Loss

To address the 75:25 class imbalance, we use Focal Loss [2]:

$$\mathcal{L}_{\text{FL}} = -\alpha\,(1 - p_t)^\gamma \log(p_t) \tag{2}$$

where p_t is the model's estimated probability for the true class, $\alpha = 0.25$, and γ controls the down-weighting of easy examples. When $\gamma = 0$, this reduces to standard cross-entropy. Higher γ values focus training more aggressively on hard, misclassified examples.

5 Training Procedure

Cross-validation. We use 5-fold Stratified Group K-Fold with subject ID as the grouping variable. This ensures zero subject overlap between train and test sets, preventing data leakage.

Preprocessing. Features are standardized per fold using training-set statistics only.

Optimization. Adam optimizer (lr= 10^{-3}, weight decay= 10^{-4}). Early stopping monitors test F1 with patience of 25 epochs. Maximum 200 epochs, batch size 32.

6 Evaluation Metrics

Given the class imbalance, we report four metrics: **F1 Score** (harmonic mean of precision and recall), **AUC-ROC** (threshold-independent discrimination), **MCC** (Matthews Correlation Coefficient, which accounts for all four confusion matrix entries [3]), and **Accuracy**.

7 Experimental Results

7.1 Baseline Performance

Table 1 shows the baseline MLP results across 5-fold subject-level CV.

Table 1: Baseline MLP [64, 32] + Focal Loss ($\gamma = 2$).

Metric	**Mean**	**Std**
Accuracy	0.831	0.057
F1 Score	0.893	0.036
AUC-ROC	0.758	0.111
MCC	0.448	0.301

These numbers are notably below the >90% accuracy reported in prior work, reflecting the impact of honest subject-level evaluation.

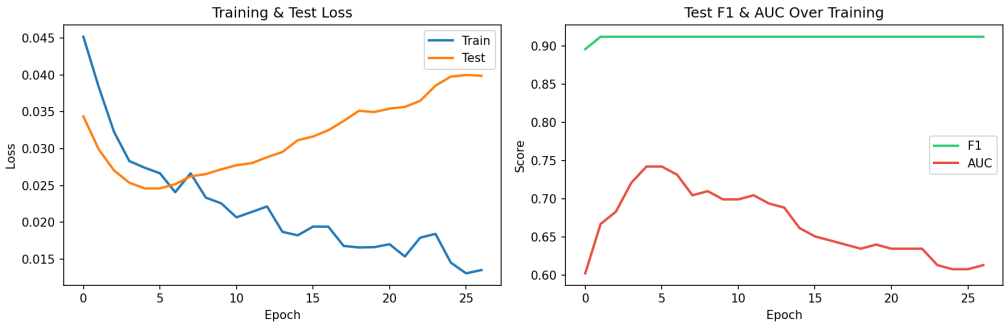

Figure 1: Training curves for the baseline model (Fold 1).

7.2 Loss Function Ablation

We compare five loss functions (Table 2).

Table 2: Loss function comparison.

Loss	**F1**	**AUC**	**MCC**	**Acc**
BCE	0.905	0.831	0.502	0.850
Weighted BCE	0.884	0.726	0.374	0.813
Focal $\gamma = 1$	0.908	0.821	0.521	0.857
Focal $\gamma = 2$	0.918	0.840	0.504	0.867
Focal $\gamma = 3$	0.903	0.802	0.492	0.851

Standard BCE and Focal Loss perform comparably on F1 ($\sim$0.90–0.92). Focal Loss with $\gamma = 2$ achieves the highest F1 (0.918) and AUC (0.840), while $\gamma = 1$ achieves the highest MCC (0.521). Weighted BCE degrades F1; this is because the code applies `pos_weight` $= 48/147 \approx 0.33$, which *down-weights* the majority positive (PD) class rather than up-weighting it, effectively worsening the imbalance instead of correcting it.

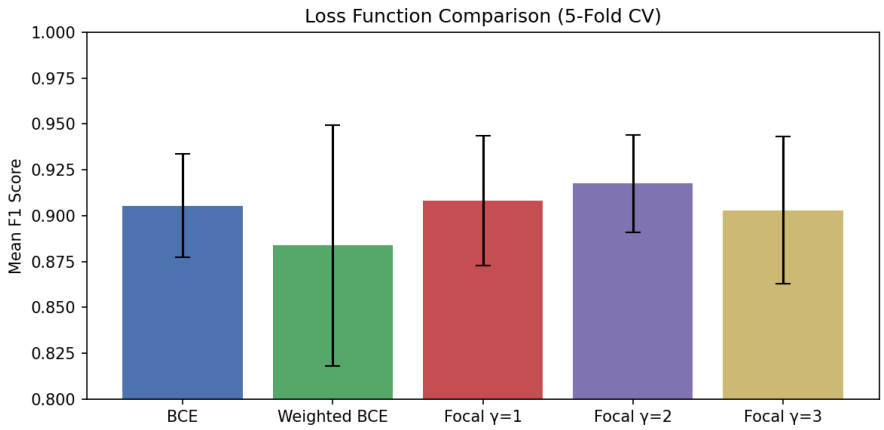

Figure 2: F1 score across loss functions.

7.3 Architecture Ablation

We compare five architectures (Table 3).

Table 3: Architecture comparison.

Name	**Hidden Layers**	**F1**	**MCC**
Shallow	[32]	0.908	0.591
Medium	[64, 32]	0.906	0.483
Deep	[128, 64, 32]	0.901	0.436
Wide	[128, 128]	0.904	0.470
Narrow-Deep	[32, 32, 32]	**0.928**	0.551

The Narrow-Deep configuration [32, 32, 32] achieves the best F1 (0.928), while the Shallow [32] model achieves the best MCC (0.591). Wider and deeper architectures do not improve performance, consistent with the small dataset size ($n = 195$) favoring lower-capacity models. Adding depth with narrow layers appears more beneficial than adding width.

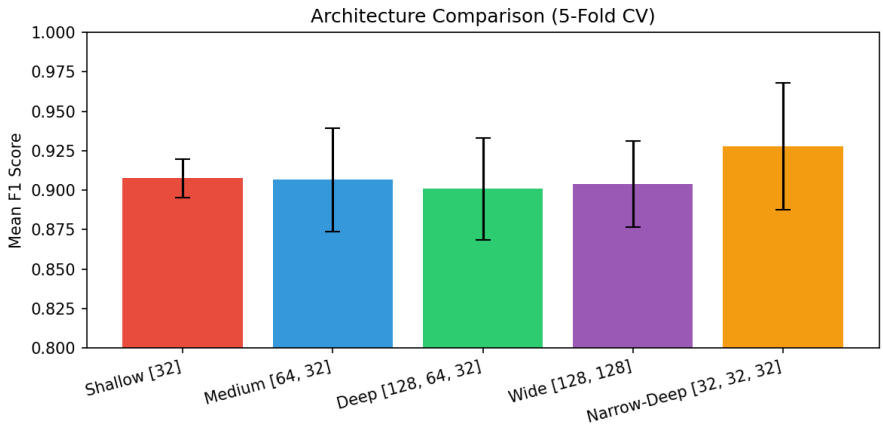

Figure 3: F1 score across architectures.

7.4 Dropout Ablation

We sweep dropout from 0.0 to 0.5 (Figure 4).

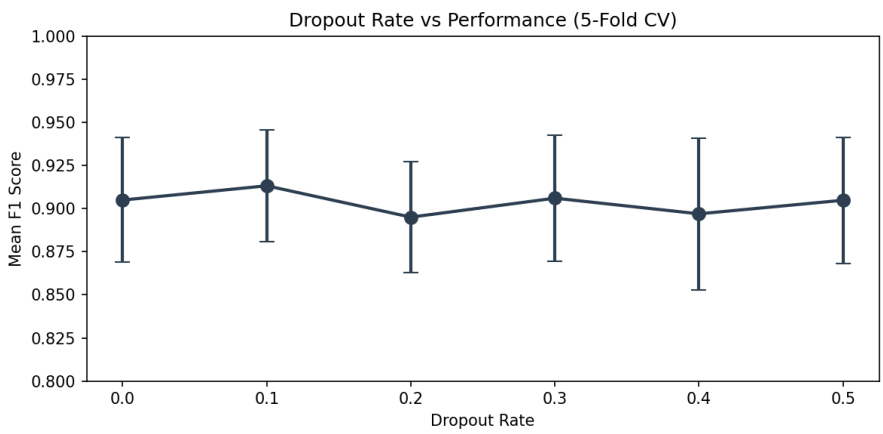

Figure 4: F1 score vs. dropout rate.

Performance peaks at $p = 0.1$ (F1=0.913, MCC=0.524), then generally decreases at higher rates. Dropout= 0.2 yields the lowest F1 (0.895), though differences across rates are modest. Even no

dropout ($p = 0.0$) performs reasonably (F1=0.905), suggesting the small dataset benefits from regularization but is not highly sensitive to the exact rate.

8 Analysis and Discussion

Subject-level evaluation matters. Our best F1 of 0.928 (Narrow-Deep [32,32,32]) and best MCC of 0.591 (Shallow [32]) are both substantially below the >90% accuracy commonly reported. This gap is attributable to subject-level splitting preventing the model from exploiting within-subject similarity.

Simpler models suffice. The narrow-deep [32, 32, 32] architecture outperforms wider alternatives, suggesting that on 195 samples, moderate capacity with appropriate regularization is optimal. This is consistent with bias-variance trade-off theory: small datasets favor lower-variance models.

High variance remains the core challenge. MCC standard deviations range from 0.13 to 0.39 across experiments. With ∼6 test subjects per fold, individual fold results are highly sensitive to which subjects are held out. Larger datasets are needed for more reliable conclusions.

Loss function choice has modest impact. All loss functions achieve F1 in the range 0.88–0.92, with the largest differences appearing in MCC and AUC. This suggests that on this dataset, the classification boundary is fairly robust to the training objective.

9 Limitations and Future Work

The primary limitation is dataset size: 195 samples from 32 subjects severely constrains model complexity and evaluation reliability. Future work could apply this framework to larger datasets (e.g., Parkinson's Telemonitoring, ∼5,875 samples), use raw audio with 1D-CNNs, or add feature interpretability methods such as Integrated Gradients or SHAP.

10 Conclusion

We presented a custom PyTorch MLP for Parkinson's disease detection that prioritizes methodological rigor: subject-level evaluation, Focal Loss for imbalance, and systematic ablation studies. Our results show that proper evaluation substantially lowers reported performance, that narrow-deep architectures outperform wider alternatives on small data, and that moderate dropout ($p \approx 0.1$) provides the best regularization. The code is fully reproducible via a single script.

11 Contribution

Wonjae Lee: Model Implementation, Training, and Evaluation Jichuan Wu: Final Report, README Leo Zheng: EDA, Visualization

References

[1] M. A. Little, P. E. McSharry, S. J. Roberts, D. A. E. Costello, and I. M. Moroz. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *BioMedical Engineering OnLine*, 6(1):23, 2007.

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[3] D. Chicco and G. Jurman. The advantages of the Matthews correlation coefficient over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21(1):6, 2020.

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GitHub Link: https://github.com/wj6801/CS273P-Final-Project